Module 3-2 Milestone 2

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The artifact I selected for my software design and engineering enhancement is the Travlr Getaways application, which I originally created in CS 465. Travlr is a full-stack web application that allows users to view travel packages and provides an administrative interface for managing trip information. The application uses Angular for the administrative interface, Node.js and Express for the API, and MongoDB with Mongoose for data storage. I selected this artifact because it demonstrates several areas of software development within one project, including frontend and backend development, database interaction, authentication, and API communication.

For my enhancement, I focused on improving the functionality and usability of the administrative side of the application. One of the main improvements was completing the CRUD functionality by adding the ability to delete trips. I added the DELETE operation throughout the application, including the Angular data service, trip component, Express route, controller, and MongoDB interaction. I also added a confirmation prompt before deletion and success and error feedback for the administrator. The DELETE API route is protected using JWT authentication so that the operation is restricted to authenticated users. During testing, I also improved the appearance and labeling of the Delete Trip button so that the destructive action is clear to the user.

I also strengthened the existing validation in the Add Trip form. The original form checked whether required fields contained values, but it did not validate whether certain values were reasonable. I added length requirements for the trip code and name, minimum values for trip length and price, and changed the appropriate fields to numeric inputs. I also added specific validation messages so users receive useful feedback when their input is invalid. Testing

confirmed that invalid information was prevented from being submitted while valid trip information could still be successfully added.

These enhancements continue to support Course Outcomes Three, Four, and Five, which I identified in my original enhancement plan. These changes show how I have improved at evaluating existing software, deciding where changes are needed, and using appropriate development tools to improve functionality while still considering security. I do not currently see a need to change the outcomes identified in my original plan because these three outcomes continue to accurately represent the skills demonstrated by the artifact.

The enhancement process gave me a better understanding of how the different layers of a full-stack application work together. One challenge was tracing the Delete Trip operation from the Angular interface through the service and API before it reached the database. I also encountered an Angular template error while adding validation messages because of incorrectly nested HTML tags. Correcting that issue and testing each change helped reinforce the importance of making incremental changes and using compiler feedback to locate problems. Overall, this enhancement made the Travlr administrative interface more complete, usable, and reliable while strengthening my understanding of full-stack software design and engineering.